Predictive Churn Analysis

Workora Labs Data Analytics Internship

A Logistic Regression classification pipeline was built to predict customer churn probability using the supplied synthetic customer churn dataset.

1. Dataset and Preprocessing

The dataset contains 800 customer records and 13 columns. The target variable is Churn. CustomerID was excluded from modeling. Categorical variables were one-hot encoded and numeric variables were MinMax scaled.

2. Train-Test Split

The data was split into 80% training data (640 records) and 20% test data (160 records), using stratification and random_state=42.

3. Model

Logistic Regression was selected as the classification model. A probability threshold of 0.50 was used to classify customers as churn or non-churn.

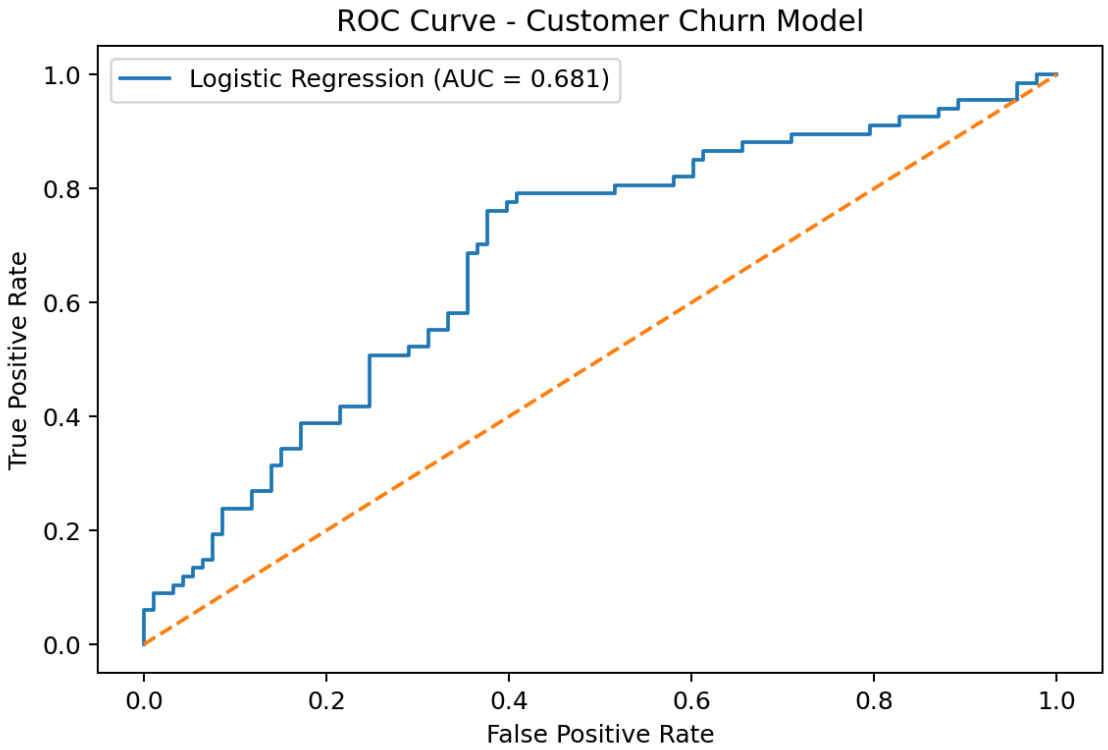
4. Model Evaluation

Metric	Test Result
Precision	0.576
Recall	0.507
F1-Score	0.540
ROC-AUC	0.681

Confusion matrix: True Negatives = 68, False Positives = 25, False Negatives = 33, True Positives = 34.

5. ROC-AUC Interpretation

The ROC-AUC is 0.681. This indicates that the model has measurable ability to distinguish customers who churn from customers who do not churn on the held-out test set. The metric should be interpreted as an evaluation of this synthetic experiment rather than a guarantee of performance on future real-world customers.



6. Churn Risk Scores

Churn probabilities were generated for all 160 test customers and exported to **customer_churn_risk_predictions.csv**. Risk levels are Low (≤33%), Medium (>33% to 66%), and High (>66%).

7. Conclusion

The Logistic Regression pipeline successfully completed preprocessing, model training, test-set evaluation, ROC-AUC analysis, and customer-level churn probability scoring. The exported risk scores can be used as a starting point for customer-retention prioritization, subject to further validation on real operational data.

Files included for submission: churn_prediction.py, customer_churn_risk_predictions.csv, churn_roc_curve.png, and this report PDF.